Module – 1 & 2

Assignment Submissions – embedUR Networking Course

29.01.2026

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1. Consider a case, a folder has multiple files and how would copy it to destination machine path (Try using SCP, cp options in Linux)

```
baala@ItsB-Laptop:~$ ip addr | grep inet
    inet 127.0.0.1/8 scope host lo
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
    inet6 ::1/128 scope host
    inet 172.19.186.187/20 brd 172.19.191.255 scope global eth0
    inet6 fe80::215:5dff:fe1d:5976/64 scope link
baala@ItsB-Laptop:~$ cd Networking
baala@ItsB-Laptop:~/Networking$ ls
Test  module1
baala@ItsB-Laptop:~/Networking$ cd Test
baala@ItsB-Laptop:~/Networking/Test$ ls
helloLinux.txt
baala@ItsB-Laptop:~/Networking/Test$
```

```
PS C:\Users\baala> scp -r C:\Users\baala\Desktop\Test baala@172.19.186.187:/home/baala/Networking/
baala@172.19.186.187's password:
helloLinux.txt                                                    100%    0     0.0KB/s   00:00
PS C:\Users\baala>
```

2. Host a FTP and SFTP server and try PUT and GET operations.

```
baala@ItsB-Laptop:~$ sudo systemctl start vsftpd
sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
baala@ItsB-Laptop:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-02 13:32:53 UTC; 3min 18s ago
   Main PID: 224 (vsftpd)
      Tasks: 1 (limit: 9354)
     Memory: 920.0K (peak: 1.5M)
        CPU: 20ms
     CGroup: /system.slice/vsftpd.service
             └─224 /usr/sbin/vsftpd /etc/vsftpd.conf

Mar 02 13:32:53 ItsB-Laptop systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Mar 02 13:32:53 ItsB-Laptop systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

```
baala@ltsB-Laptop:~$ ftp 172.19.186.187
Connected to 172.19.186.187.
220 (vsFTPd 3.0.5)
Name (172.19.186.187:baala): baala
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||25671|)
150 Here comes the directory listing.
drwx------    2 1000     1000         4096 Feb 04 15:50 Downloads
-rw-r--r--    1 1000     1000           16 Mar 02 05:42 FTPCheck.txt
drwxr-xr-x    4 1000     1000         4096 Mar 02 05:31 Networking
drwxr-xr-x    7 1000     1000         4096 Feb 05 14:02 assignments
-rw-r--r--    1 1000     1000          165 Feb 01 06:49 binLargeFiles.txt
-rw-r--r--    1 1000     1000           52 Feb 17 16:39 error.log
-rwxr-xr-x    1 1000     1000           68 Feb 16 16:31 hehe.txt
-rwxr-xr-x    1 1000     1000           97 Feb 17 14:00 hehee.sh
lrwxrwxrwx    1 1000     1000           18 Feb 14 16:45 hellolink.sh -> shellprac/hello.sh
-rwxr-xr-x    1 1000     1000          152 Feb 17 14:03 pass.sh
-rwxr-xr-x    1 1000     1000           92 Feb 12 14:05 script.sh
drwxr-xr-x    2 1000     1000         4096 Feb 14 10:39 shellprac
-rw-r--r--    1 1000     1000           16 Mar 02 05:39 test.txt
226 Directory send OK.
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||34814|)
150 Ok to send data.
  0% |                                                                    |     0       0.00 KiB/s    --:-- ETA
226 Transfer complete.
ftp> get test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||58756|)
150 Opening BINARY mode data connection for test.txt (0 bytes).
     0        0.00 KiB/s
226 Transfer complete.
ftp> ls
229 Entering Extended Passive Mode (|||62267|)
```

```
ftp> ls
229 Entering Extended Passive Mode (|||62267|)
150 Here comes the directory listing.
drwx------    2 1000     1000         4096 Feb 04 15:50 Downloads
-rw-r--r--    1 1000     1000           16 Mar 02 05:42 FTPCheck.txt
drwxr-xr-x    4 1000     1000         4096 Mar 02 05:31 Networking
drwxr-xr-x    7 1000     1000         4096 Feb 05 14:02 assignments
-rw-r--r--    1 1000     1000          165 Feb 01 06:49 binLargeFiles.txt
-rw-r--r--    1 1000     1000           52 Feb 17 16:39 error.log
-rwxr-xr-x    1 1000     1000           68 Feb 16 16:31 hehe.txt
-rwxr-xr-x    1 1000     1000           97 Feb 17 14:00 hehee.sh
lrwxrwxrwx    1 1000     1000           18 Feb 14 16:45 hellolink.sh -> shellprac/hello.sh
-rwxr-xr-x    1 1000     1000          152 Feb 17 14:03 pass.sh
-rwxr-xr-x    1 1000     1000           92 Feb 12 14:05 script.sh
drwxr-xr-x    2 1000     1000         4096 Feb 14 10:39 shellprac
-rw-r--r--    1 1000     1000            0 Mar 02 13:38 test.txt
226 Directory send OK.
ftp> |
```

```
PS C:\Users\baala> sftp baala@172.19.186.187
baala@172.19.186.187's password:
Connected to 172.19.186.187.
sftp> ls
Downloads          FTPCheck.txt       Networking         assignments        binLargeFiles.txt  error.log          hehe.txt
hehee.sh           hellolink.sh       pass.sh            script.sh          shellprac          test.txt
sftp> put baala.txt
Uploading baala.txt to /home/baala/baala.txt
baala.txt                                                                                   100%   42     0.0KB/s   00:00
sftp> get test.txt
Fetching /home/baala/test.txt to test.txt
sftp> |
```

3. Explore with Wireshark/TCP-dump/cisco packet tracer tools and learn about packets filters.

ip.addr == 172.19.186.187

No.	Time	Source	Destination	Protocol	Length	Info
→ 246	9.311376	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=1/256, ttl=64 (reply in 247)
← 247	9.314677	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=1/256, ttl=114 (request in 246)
267	10.318955	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=2/512, ttl=64 (reply in 268)
268	10.321598	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=2/512, ttl=114 (request in 267)
309	11.327551	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=3/768, ttl=64 (reply in 310)
310	11.331023	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=3/768, ttl=114 (request in 309)
319	12.336632	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=4/1024, ttl=64 (reply in 320)
320	12.339892	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=4/1024, ttl=114 (request in 319)
335	13.345861	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=5/1280, ttl=64 (reply in 336)
336	13.348630	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=5/1280, ttl=114 (request in 335)
434	14.356186	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=6/1536, ttl=64 (reply in 435)
435	14.359054	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=6/1536, ttl=114 (request in 434)
449	15.366874	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=7/1792, ttl=64 (reply in 450)
450	15.370292	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=7/1792, ttl=114 (request in 449)
491	16.377505	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=8/2048, ttl=64 (reply in 492)
492	16.403989	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=8/2048, ttl=114 (request in 491)
551	17.387659	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=9/2304, ttl=64 (reply in 552)
552	17.390504	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=9/2304, ttl=114 (request in 551)
572	18.399281	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=10/2560, ttl=64 (reply in 574)
574	18.422990	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=10/2560, ttl=114 (request in 572)
584	19.411016	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=11/2816, ttl=64 (reply in 585)
585	19.414079	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=11/2816, ttl=114 (request in 584)
597	20.423475	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=12/3072, ttl=64 (reply in 598)
598	20.427142	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=12/3072, ttl=114 (request in 597)
613	21.435523	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=13/3328, ttl=64 (reply in 614)
614	21.438362	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=13/3328, ttl=114 (request in 613)
623	22.447322	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=14/3584, ttl=64 (reply in 624)
624	22.450282	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=14/3584, ttl=114 (request in 623)
633	23.459903	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=15/3840, ttl=64 (reply in 634)

icmp

No.	Time	Source	Destination	Protocol	Length	Info
239	9.311435	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=1/256, ttl=63 (reply in 240)
240	9.314588	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=1/256, ttl=115 (request in 239)
→ 246	9.311376	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=1/256, ttl=64 (reply in 247)
← 247	9.314677	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=1/256, ttl=114 (request in 246)
263	10.318996	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=2/512, ttl=63 (reply in 264)
264	10.321553	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=2/512, ttl=115 (request in 263)
267	10.318955	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=2/512, ttl=64 (reply in 268)
268	10.321598	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=2/512, ttl=114 (request in 267)
304	11.327597	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=3/768, ttl=63 (reply in 305)
305	11.330968	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=3/768, ttl=115 (request in 304)
309	11.327551	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=3/768, ttl=64 (reply in 310)
310	11.331023	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=3/768, ttl=114 (request in 309)
319	12.336632	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=4/1024, ttl=64 (reply in 320)
320	12.339892	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=4/1024, ttl=114 (request in 319)
321	12.336708	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=4/1024, ttl=63 (reply in 322)
322	12.339834	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=4/1024, ttl=115 (request in 321)
333	13.345909	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=5/1280, ttl=63 (reply in 334)
334	13.348569	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=5/1280, ttl=115 (request in 333)
335	13.345861	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=5/1280, ttl=64 (reply in 336)
336	13.348630	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=5/1280, ttl=114 (request in 335)
402	13.681086	172.20.181.3	172.20.72.11	ICMP	295	Destination unreachable (Port unreachable)
434	14.356186	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=6/1536, ttl=64 (reply in 435)
435	14.359054	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=6/1536, ttl=114 (request in 434)
440	14.356246	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=6/1536, ttl=63 (reply in 441)
441	14.359005	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=6/1536, ttl=115 (request in 440)
449	15.366874	172.19.186.187	142.250.66.14	ICMP	98	Echo (ping) request id=0x0669, seq=7/1792, ttl=64 (reply in 450)
450	15.370292	142.250.66.14	172.19.186.187	ICMP	98	Echo (ping) reply id=0x0669, seq=7/1792, ttl=114 (request in 449)
453	15.366916	172.20.181.3	142.250.66.14	ICMP	98	Echo (ping) request id=0x03e8, seq=7/1792, ttl=63 (reply in 454)
454	15.370230	142.250.66.14	172.20.181.3	ICMP	98	Echo (ping) reply id=0x03e8, seq=7/1792, ttl=115 (request in 453)

```
baala@ItsB-Laptop:~$ sudo tcpdump -i eth0
[sudo] password for baala:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
Listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:51:44.529045 IP 172.19.186.187.46007 > prod-ntp-3.ntp4.ps5.canonical.com.ntp: NTPv4, Client, length 48
13:51:44.699809 IP prod-ntp-3.ntp4.ps5.canonical.com.ntp > 172.19.186.187.46007: NTPv4, Server, length 48
13:51:49.969689 ARP, Request who-has 172.19.186.187 (00:15:5d:1d:50:d3 (oui Unknown)) tell ItsB-Laptop.mshome.net, length 28
13:51:49.969719 ARP, Reply 172.19.186.187 is-at 00:15:5d:1d:50:d3 (oui Unknown), length 28
13:51:50.247593 ARP, Request who-has ItsB-Laptop.mshome.net tell 172.19.186.187, length 28
13:51:50.248278 ARP, Reply ItsB-Laptop.mshome.net is-at 00:15:5d:7c:ae:0f (oui Unknown), length 28
13:52:17.284685 IP 172.19.186.187.46925 > prod-ntp-3.ntp4.ps5.canonical.com.ntp: NTPv4, Client, length 48
13:52:17.458104 IP prod-ntp-3.ntp4.ps5.canonical.com.ntp > 172.19.186.187.46925: NTPv4, Server, length 48
13:52:23.017630 ARP, Request who-has ItsB-Laptop.mshome.net tell 172.19.186.187, length 28
13:52:23.017983 ARP, Reply ItsB-Laptop.mshome.net is-at 00:15:5d:7c:ae:0f (oui Unknown), length 28
13:52:50.048199 IP 172.19.186.187.58539 > prod-ntp-3.ntp4.ps5.canonical.com.ntp: NTPv4, Client, length 48
13:52:50.210137 IP prod-ntp-3.ntp4.ps5.canonical.com.ntp > 172.19.186.187.58539: NTPv4, Server, length 48
13:52:55.480161 ARP, Request who-has 172.19.186.187 (00:15:5d:1d:50:d3 (oui Unknown)) tell ItsB-Laptop.mshome.net, length 28
13:52:55.480181 ARP, Reply 172.19.186.187 is-at 00:15:5d:1d:50:d3 (oui Unknown), length 28
13:52:55.808172 ARP, Request who-has ItsB-Laptop.mshome.net tell 172.19.186.187, length 28
13:52:55.808534 ARP, Reply ItsB-Laptop.mshome.net is-at 00:15:5d:7c:ae:0f (oui Unknown), length 28
13:53:20.464691 IP 0.0.0.0 > 255.255.255.255:  hip 68
13:53:22.832688 IP 172.19.186.187.34505 > prod-ntp-3.ntp4.ps5.canonical.com.ntp: NTPv4, Client, length 48
13:53:23.021946 IP prod-ntp-3.ntp4.ps5.canonical.com.ntp > 172.19.186.187.34505: NTPv4, Server, length 48
13:53:28.432428 ARP, Request who-has 172.19.186.187 (00:15:5d:1d:50:d3 (oui Unknown)) tell ItsB-Laptop.mshome.net, length 28
13:53:28.432443 ARP, Reply 172.19.186.187 is-at 00:15:5d:1d:50:d3 (oui Unknown), length 28
13:53:28.566938 ARP, Request who-has ItsB-Laptop.mshome.net tell 172.19.186.187, length 28
13:53:28.567422 ARP, Reply ItsB-Laptop.mshome.net is-at 00:15:5d:7c:ae:0f (oui Unknown), length 28
^C
23 packets captured
23 packets received by filter
0 packets dropped by kernel
baala@ItsB-Laptop:~$
```

4. Understand linux utility commands like - ping, arp (Understand each params from ifconfig output)

ping checks if the host is reachable and latency to the url posted. It uses ICMP protocol.

```
baala@ItsB-Laptop:~$ ping -c 10 google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=114 time=3.57 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=114 time=2.83 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=114 time=4.37 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=114 time=3.62 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=114 time=2.99 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=6 ttl=114 time=3.48 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=7 ttl=114 time=2.90 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=8 ttl=114 time=2.90 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=9 ttl=114 time=17.5 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=10 ttl=114 time=3.22 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9145ms
rtt min/avg/max/mdev = 2.834/4.736/17.481/4.271 ms
```

ARP – address resolution protocol

```
baala@ItsB-Laptop:~$ arp -a
ItsB-Laptop.mshome.net (172.19.176.1) at 00:15:5d:7c:ae:0f [ether] on eth0
baala@ItsB-Laptop:~$ ip neigh
172.19.176.1 dev eth0 lladdr 00:15:5d:7c:ae:0f STALE
baala@ItsB-Laptop:~$
```

Ip addr

```
baala@ItsB-Laptop:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:1d:50:d3 brd ff:ff:ff:ff:ff:ff
    inet 172.19.186.187/20 brd 172.19.191.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe1d:50d3/64 scope link
       valid_lft forever preferred_lft forever
```

Ethtool

```
baala@ItsB-Laptop:~$ sudo ethtool eth0
Settings for eth0:
        Supported ports: [  ]
        Supported link modes:   Not reported
        Supported pause frame use: No
        Supports auto-negotiation: No
        Supported FEC modes: Not reported
        Advertised link modes:  Not reported
        Advertised pause frame use: No
        Advertised auto-negotiation: No
        Advertised FEC modes: Not reported
        Speed: 10000Mb/s
        Duplex: Full
        Auto-negotiation: off
        Port: Other
        PHYAD: 0
        Transceiver: internal
        Current message level: 0x000000f7 (247)
                               drv probe link ifdown ifup rx_err tx_err
        Link detected: yes
```

5. Understand what happens when duplicate IPs configured in a network.

   When duplicate IP addresses are configured in the same network, two devices attempt to use the same identifier, causing IP conflict. It leads to ARP table instability where the IP address maps to different MAC addresses causing intermittent connectivity issues, packet loss, slow performance, or complete network disconnection in some cases.

6. Understand how to access remote system using (VNCviewer, Anydesk, teamviewer and remote desktop connections)

   When accessing a remote system, tools like VNC Viewer, AnyDesk, TeamViewer, and Remote Desktop Connection allow users to control another computer over a network or the internet. VNC Viewer works using the VNC protocol and requires the VNC server to be installed on the remote machine. AnyDesk and TeamViewer are third-party remote access applications that provide secure, encrypted connections using a unique ID and password, commonly used for remote support. Remote Desktop Connection (RDP) is a built-in Windows feature that allows users to connect to another Windows system using its IP address or hostname, provided Remote Desktop is enabled. All these tools send screen, keyboard, and mouse data over the network, enabling full remote control of the target system.

7. How to check your default gateway is reachable or not and understand about default gateway.

   The default gateway is the usually a router that forwards traffic from a local network to other networks. To check if the default gateway is reachable, it is identified it using the command ip route (Linux) or ipconfig (Windows), where it appears as “default via” or “Default Gateway.” Then, the ping command followed by the gateway’s IP address is used to test connectivity. If the gateway responds, it means system can communicate with the router else there may be issues such as incorrect IP configuration, cable disconnection, Wi-Fi problems, or router failure.

8. Check iwconfig/ifconfig to understand in detail about network interfaces (check about interface speed, MTU and other parameters)

   These commands display important parameters such as the interface name (e.g., eth0 or wlan0), IP address, subnet mask, and MAC address (link/ether). They also show the MTU (Maximum Transmission Unit), which defines the maximum packet size that can be transmitted, typically 1500 bytes for Ethernet. The output includes interface status flags like UP (enabled), BROADCAST, MULTICAST, and RUNNING, indicating whether the interface is active. Using tools like ethtool, we can check interface speed (e.g., 100Mbps or 1Gbps) and duplex mode (full/half).

9. Log in to your home router’s web interface (usually at 192.168.1.1 or 192.168.0.1) and check the connected devices list.

   This list had each device’s IP address, MAC address, and hostname, helping monitor network activity and identify unknown devices. (I reside in hostel and it only allows access to my network usage statistics)

10. Explain how a DHCP server assigns IP addresses to devices in your network.

    A DHCP (Dynamic Host Configuration Protocol) server automatically assigns IP addresses to devices in a network to avoid manual configuration and IP conflicts. When a device connects to the network, it sends a DHCP Discover message to request an IP address. The DHCP server responds with a DHCP Offer containing an available IP address, subnet mask, default gateway, DNS server, and lease time. The device then sends a DHCP Request to accept the offer, and the server completes the process with a DHCP Acknowledgment (ACK). This four-step process (Discover, Offer, Request, Acknowledge) ensures each device receives a unique IP address for a specific lease period, after which it must renew the address.

11. Using a terminal, connect to a remote machine via SSH and telnet.

```
PS C:\Users\baala> ssh baala@172.19.186.187
baala@172.19.186.187's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Mar  2 15:11:08 UTC 2026

  System load:  0.0                 Processes:             33
  Usage of /:   0.4% of 1006.85GB   Users logged in:       1
  Memory usage: 5%                  IPv4 address for eth0: 172.19.186.187
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Sun Mar  1 11:01:13 2026 from 172.19.176.1
baala@ItsB-Laptop:~$ sudo apt install telnet -y
[sudo] password for baala:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
telnet is already the newest version (0.17+2.5-3ubuntu4.1).
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
baala@ItsB-Laptop:~$
```